Vibration Measurement Using Accelerometer Sensor and Fast Fourier Transform

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Abstract Vibration monitoring is necessary for any system for its effective performance, product quality, safety and life span. It helps to diagnose the health of system for any predictive maintenance needed. This paper presents the analysis of vibration signal using fast Fourier transform (FFT). Frequency spectrum of vibrating source gives information of the vibration level. The natural frequency of the vibrating source was found to be 17 Hz with total harmonic distortion (THD) of 0.000177%. The analysis was carried out using NI LabVIEW software.

Keywords Vibration · Sensor · Accelerometer · FFT · LabVIEW

1 Introduction

Vibration can be described as the periodic motion in alternately opposite directions about a reference equilibrium position. For proper functioning of a plant or system, it is very important that all machines should work properly. Vibration is one of the key parameters to measure and analyze constantly for good quality product and safety. It also helps in diagnosing the health of the system and predictive maintenance. Routine monitoring of vibration is necessary to avoid any failures. Vibration measurements are considered as a part of performance test for instrument while in use. It gives an indication of how well the instrument or system has been designed and manufactured and can also provide advanced warning of possible operational problems.

There are various factors influencing vibration measurements which include cross-axis sensitivity, capacitance effect, sensor loading, coupling and other external factors such as temperature and magnetic field.

A body is said to vibrate when it oscillates about a reference point. Vibration is caused by mechanical disturbance from some source and is transferred to the system in contact with it. The need of vibration measurement is required due to growth of environmental testing and health of the system. It can be expressed in terms

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of displacement, velocity or acceleration. The analysis, measurement and testing of vibration are usually done for the rotating or reciprocating instruments. All the system or machine performance should withstand a particular level of vibration for effective running of the process. If the system is not able to withstand the vibration of the instrument, then it may collapse or lead to complete failure of the system. It is not possible to make completely perfect system as per design or mathematical model. Vibration detection is applicable in the working of machines, construction of buildings and bridges, structural health monitoring, security reasons, forecasting of natural disasters such as earthquake and tsunami.

Most vibrations are sinusoidal in nature about its mean position. The quantities required to be measured in a vibrating system are displacement (x), velocity (v) and acceleration (a) expressed in Eqs. (1), (2) and (3), respectively. Sinusoidal vibrations are expressed in terms of amplitude and frequency or maximum velocity (v_0) or maximum acceleration (a_0) in Eqs. 4 and 5, respectively.

$$x = x_m \sin \omega t \tag{1}$$

where x_m is amplitude, and ω is angular frequency in terms of rad/s.

$$v = \dot{x} = x_m \omega \cos \omega t \tag{2}$$

$$a = \ddot{x} = x_m \omega^2 \tag{3}$$

$$v_0 = x_m \omega \tag{4}$$

$$a_0 = -x_m \omega^2 \tag{5}$$

Any sensor which is sensitive to the amplitude (displacement), velocity or acceleration can be used to measure vibration. Seismic transducer is the device to measure the same which can be used in two different modes, i.e., displacement mode and acceleration mode. There are other types of vibration sensors available such as inductive sensor, capacitive sensor, piezoelectric sensor, magnetic sensor, optical fiber sensor and photoelectric sensor. Accelerometer is the most common sensor used for vibration analysis and data collection. Many accelerometers are available based on their working principles such as capacitive accelerometer, piezoelectric accelerometer and hall effect accelerometer.

Mathematical model of vibration in matrix form can be represented as equation of motion as given in Eq. (6). M, C and K represent inertia matrix, damping matrix and stiffness matrix, respectively, where x is position vector and F is input vector.

$$M\ddot{x} + C\dot{x} + Kx = F(t) \tag{6}$$

Various works in the field of vibration are done by many researchers worldwide. Song et al. [1] and team have explained uncertainties of measurement of vibration in terms of acceleration. Usuda and Imai [2] explained a different prospect of vibration activity measurement using IMEKO TC22. Chen et al. [3] designed an optical fiber-based model for vibration detection. It was based on the changes in geometric curvature of the optical fiber with respect to any vibration. Uchimura et al. [4] have described wireless sensing of vibration through IEEE 802.11-based TSF counter. Igor Kurytnik [5] and team have proposed ZigBee sensor network for detection of vibration. Sivakumar [6] has simulated the random vibration analysis of complete aircraft for active and passive gears. It was found that performance and life of aircraft was improved with active gears. A mathematical model of moving vehicle is simulated by Zhou and Qiu [7], and vehicle performance was mostly effected by seat vibration and engine vibration. Zhang et al. [8] have analyzed vibration using finite element method and FFT. Sabato [9] has monitored pedestrian vibration using wireless MEMS accelerometer board. Wada et al. [10] have sensed multipoint vibration using FBG and current modulated laser diode. A new cost-effective vibration sensor is developed based on micro-wire sensor system using FFT [11]. Using SCILAB simulation software, mathematical modeling and analysis of vibration were done for rotating machinery [12]. Maruthi et al. [13] have done the mathematical analysis of unbalanced magnetic pull and detection of mixed air gap eccentricity in induction motor by vibration analysis using MEMS accelerometer. Hu Jingjing [14] and team have studied the equivalent simplified model of multilayer vertical isolation structure under heavy load train vibration. The structure is based on the equivalent principle of the first two order frequency and the total axial force.

2 Experimental Setup

The experiment setup, shown in Fig. 1, consists of a vibrating source, accelerometer sensor, pulse analyzer and a computer to display the output. Sensor module consists of a vibrating device to generate vibration, magnetic dart and an accelerometer as sensor. An accelerometer is attached to the vibrating source to pick up the vibration. Pulse analyzer is used as signal processing unit. Figure 2 shows the block diagram of the complete measurement setup. The pulse analyzer (OR38, OROS 3-Series/NVGate) receives signal from the accelerometer. The detected vibration output is in terms of acceleration (m/s^2) with respect to frequency (Hz) with the interval of 50 Hz. Figure 3 shows the output received from pulse analyzer on the computer.

The output data taken from pulse analyzer are processed using fast Fourier transform in National Instruments (NI) LabVIEW software.

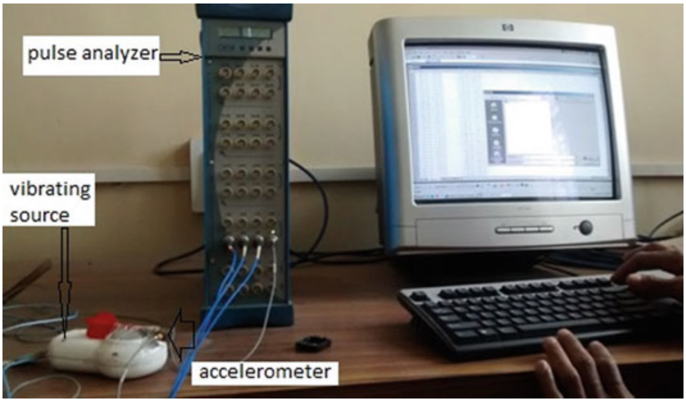

Fig. 1 Vibration measurement setup includes a vibrating source, accelerometer, pulse analyzer and computer

Fig. 2 Block diagram of vibration measurement setup

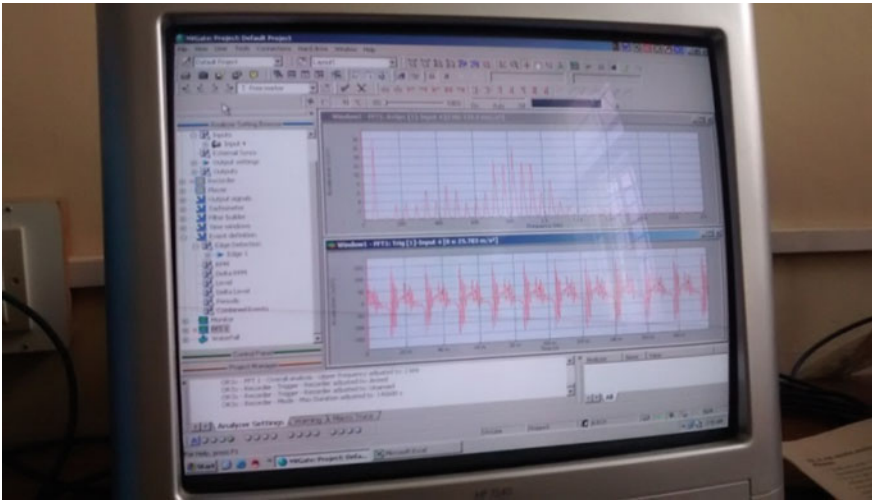

Fig. 3 Pulse analyzer output display in computer

3 Result and Discussion

Figures 4 and 5 show the vibration spectrum analysis waveforms (output) received from pulse analyzer in terms of acceleration (m/s^2) with respect to change in frequency (Hz) from 0 to maximum of 2and 20 kHz frequency, respectively. The experiment was carried out for different frequency range. An efficient algorithm, i.e., fast Fourier transform (FFT) is used for computation of data. The Fourier transform converts output of vibration amplitude as a function of frequency so that the analyzer can detect the source the vibration. Figure 6 represents frequency (Hz) which is plotted with respect to acceleration (m/s^2) using NI LabVIEW software. The maximum acceleration was observed at 17 Hz frequency.

For the purpose of spectrum analysis, the conventional discrete Fourier transform (DFT) is one of the popular tools [15]. It gives satisfactory result for the vibration under the stationary conditions. The advent of fast Fourier transform (FFT) has made spectra measurement easier and more efficient. Figures 7 and 8 show that the FFT results for the vibration signal are acquired and analyzed using FFT. The data received were received by pulse generator, and figures show the FFT in magnitude and phase, respectively. The frequency spectrum represents the total amplitude at each of these frequencies; it is calculated as the square root of the sum of the squares of the coefficients of the sine and cosine components. The fundamental frequency

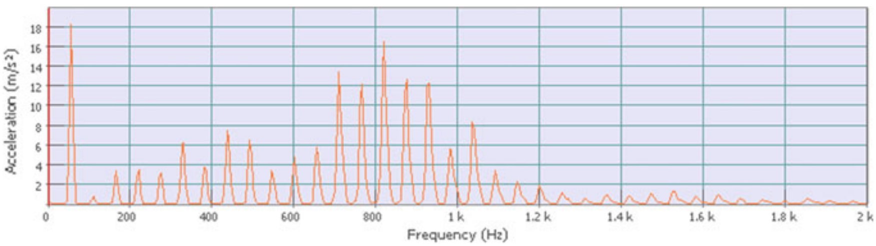

Fig. 4 Vibration spectrum analysis: acceleration (m/s^2) vs frequency (Hz) plot generated by pulse analyzer in the range of 0–2 kHz frequency

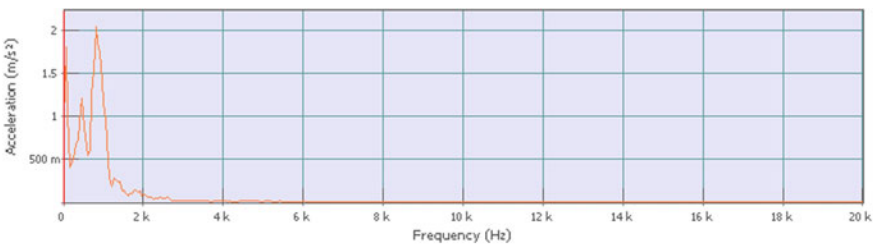

Fig. 5 Vibration spectrum analysis: acceleration (m/s^2) versus frequency (Hz) plot generated by pulse analyzer ranging from 0 to 20 kHz frequency

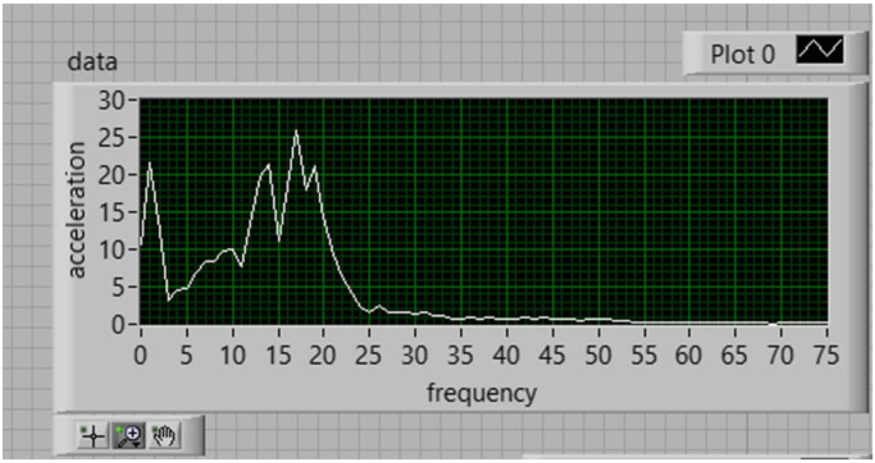

Fig. 6 Vibration spectrum analysis: frequency (Hz) versus acceleration (m/s^2) plot in NI LabVIEW software

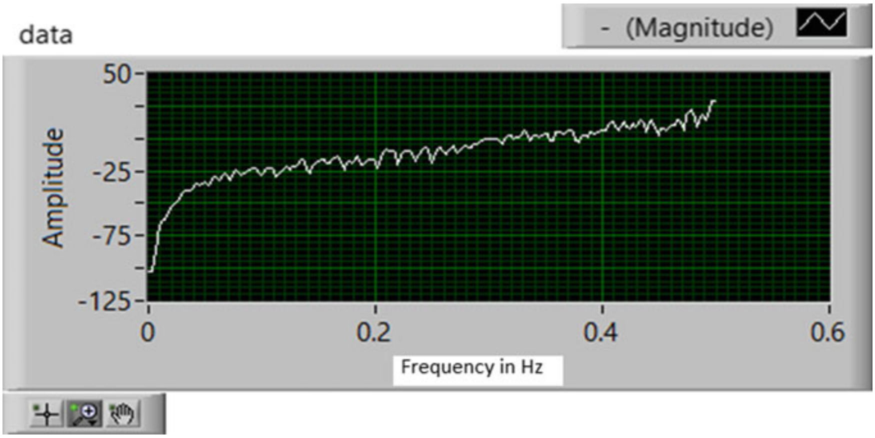

Fig. 7 FFT analysis (amplitude) using NI LabVIEW software

measured is 17 Hz, and the second highest peak appears at 850 Hz. 0.000177% total harmonic distortion (TDH) was calculated.

4 Conclusion

This paper shows the vibration spectrum analysis using fast Fourier transform. The measurement and frequency analyses of the vibration signal are measured from a

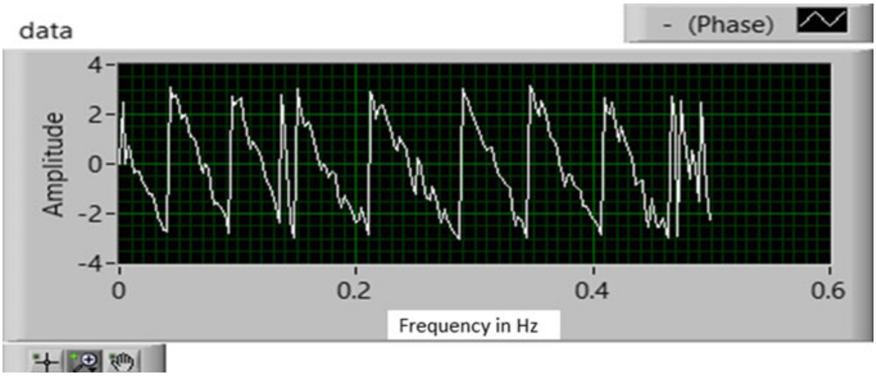

Fig. 8 FFT analysis (phase) using NI LabVIEW software

vibrating source. Vibration was picked by the accelerometer and sent to pulse generator for signal processing. Natural frequency of vibrating source for the setup was calculated as 17 Hz. Second highest peak was observed at 850 Hz with 0.000177% THD.

5 Future Scope

A new optical method for vibration detection can be explored using magneto-optic sensor. This will make the sensor fast and accurate without electrical and environmental interference.

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